

# ENSO-Aware Adversarial Forecasting: A Unified WGAN-GP Framework for Extreme-Aware Imputation and Forecasting of Heatwave Days in Regional Temperature Time Series

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**Abstract**—Weather and climate time series are simultaneously non-stationary, gap-prone, and characterized by rare but high-impact extreme events such as heatwaves. Over India, the frequency and persistence of heatwaves are further modulated by the El Niño–Southern Oscillation (ENSO), so that the days of greatest practical consequence cluster in El Niño-affected seasons. Existing generative adversarial network (GAN) approaches to this domain address imputation and forecasting as separate problems and ignore large-scale climate drivers such as ENSO. We propose a unified adversarial framework in which a single ENSO-conditioned generator performs both missing-data imputation and multi-step forecasting of heatwave days for a regional temperature series, trained with a Wasserstein GAN with gradient penalty (WGAN-GP) objective augmented by a generalized-extreme-value-inspired exceedance-weighted loss term. The generator and critic are conditioned on a causally lagged Oceanic Niño Index, so that the model can learn how the likelihood and intensity of heatwave days shift with the ENSO phase. The framework is evaluated on ERA5-Land daily reanalysis for Chennai, India, under a temporally held-out protocol, with continuous-value metrics (RMSE, MAE,  $R^2$ , extreme-tail RMSE) and event-level metrics (probability of detection, false alarm ratio, critical success index) aligned to the India Meteorological Department heatwave definition and reported separately for El Niño, La Niña and neutral periods.

**Index Terms**—generative adversarial networks, WGAN-GP, time series imputation, weather forecasting, extreme value theory, heatwave prediction, El Niño–Southern Oscillation, ERA5-Land

## I. INTRODUCTION

Missing observations and the need to project future states are two persistent, related challenges in regional climate time series analysis. Sensor outages, transmission loss, and irregular station coverage all produce gaps that must be filled before a series can be analyzed or modeled, while early-warning systems require genuine forecasts of future values. Both problems are compounded when the quantity of interest is prone to extreme excursions: temperature series in heatwave-prone regions such as peninsular India are dominated in aggregate by smooth seasonal cycles, yet the events of greatest practical

consequence—heatwaves—sit in the tail of the distribution, precisely where models trained under a mean-squared-error objective are least reliable.

Heatwave occurrence over India is not stationary from year to year. ENSO, the dominant mode of interannual climate variability in the tropical Pacific [1], [2], influences Indian surface temperature through atmospheric teleconnections, and the frequency and duration of Indian heatwaves have been linked to El Niño conditions in the central and eastern Pacific [3], [4]. A model that imputes or forecasts temperature without access to the ENSO state must treat El Niño and La Niña years as draws from a single distribution, and is therefore likely to under-predict exactly the heatwave days that cluster in El Niño-affected seasons.

GANs [5] have been applied to both imputation [6] and forecasting [7] of weather data, and extreme-value-aware objectives improve tail fidelity in both non-adversarial [8] and adversarial [9]–[11] settings. However, no existing framework combines imputation and forecasting within a single adversarially-trained generator for a region-scale series, no extreme-value-aware GAN has been applied to city-scale temperature in a heatwave-prone setting such as India, and none conditions on the ENSO state that modulates heatwave risk. A closely related study on Chennai heatwave classification [12] names time series GANs as a plausible, unexplored strategy for the class imbalance of rare heatwave events, but does not pursue it.

The proposed work makes the following contributions:

- 1) A unified adversarial framework in which a single generator performs both interior-gap imputation and multi-step forecasting, differing only in where the reconstruction mask falls within a training window.
- 2) ENSO conditioning of both the generator and the WGAN-GP critic on a causally lagged Oceanic Niño Index (ONI), so that imputed and forecast heatwave days are generated conditionally on the El Niño, La Niña or neutral state of the tropical Pacific.

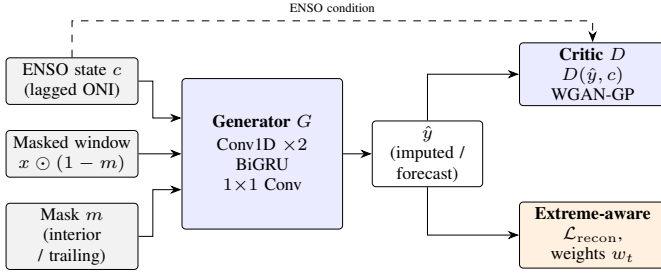


Fig. 1. Proposed ENSO-aware framework. A single generator receives a masked temperature window, the mask, and the lagged ENSO state, and fills either an interior gap (imputation) or the trailing  $k$  days (forecasting). It is trained with an ENSO-conditioned WGAN-GP critic and an extreme-aware weighted reconstruction loss.

- 3) A generalized-extreme-value-inspired exceedance-weighted loss that makes the reconstruction objective explicitly sensitive to heatwave behavior, combined with a WGAN-GP adversarial objective [13], [14].
- 4) A temporally held-out evaluation protocol that reports continuous and event-level (POD/FAR/CSI) metrics stratified by ENSO phase, for direct comparability with classification-style heatwave studies.

## II. PROPOSED METHODOLOGY

### A. Data

Daily 2 m air temperature and auxiliary variables are taken from the ERA5-Land reanalysis [15] via Google Earth Engine [16] for Chennai (13.08 °N, 80.23 °E) and split chronologically into training, validation and test periods; scaling parameters and thresholds are fit on the training period only. The ENSO state is the monthly NOAA ONI [17], the three-month running mean of ERSSTv5 Niño-3.4 SST anomalies [18]; months are labeled El Niño (La Niña) when the ONI is at or above +0.5 °C (at or below −0.5 °C) for at least five consecutive overlapping seasons. Because the ONI is a centered mean, the model uses only the value published before the start of each window (a lag of  $\ell \geq 2$  months), keeping the forecasts operationally causal. Heatwave days follow the India Meteorological Department criterion for coastal stations [19]: a departure of at least 4.5 °C from the calendar-day normal of the daily maximum, with an actual maximum of at least 37 °C.

### B. Unified Masked Reconstruction

Given a window of length  $W$  and a binary mask  $m \in \{0, 1\}^W$ , the generator fills the hidden positions. An interior masked span is an imputation task; a trailing span of  $k$  steps is a  $k$ -step forecast. The mask type, position and length are resampled for every training window, so that the model is never trained on the exact segments it is scored on. Each window carries a conditioning vector  $c$  holding the lagged ONI, and the generator learns  $p(y_{\mathcal{M}} | y_{\mathcal{N}}, c)$ , the distribution of the hidden values given the observed context and the ENSO state.

### C. Architecture and Adversarial Objective

The generator consists of two 1-D convolutional layers (kernel size 5) over the masked values, the mask indicator and the ENSO channels, followed by a bidirectional GRU and a  $1 \times 1$  convolution. The critic is convolutional and conditional [20]: it scores a completed window together with  $c$ , so that it judges whether a heatwave sequence is realistic for the given ENSO state. It is trained with the WGAN-GP loss [14]

$$\mathcal{L}_D = \mathbb{E}[D(\hat{x}, c)] - \mathbb{E}[D(x, c)] + \gamma \mathbb{E}\left[\left(\|\nabla_{\tilde{x}} D(\tilde{x}, c)\|_2 - 1\right)^2\right], \quad (1)$$

where  $\hat{x}$  is the generator-completed window and  $\tilde{x}$  a random interpolation between real and completed windows.

### D. Extreme-Value-Aware Loss

Following the extreme value loss of Ding et al. [8], masked positions are weighted by their exceedance over the heatwave threshold  $\tau_t$ :

$$w_t = 1 + \lambda \cdot \max\left(0, \frac{y_t - \tau_t}{\sigma}\right), \quad (2)$$

$$\mathcal{L}_{\text{recon}} = \frac{1}{|\mathcal{M}|} \sum_{t \in \mathcal{M}} w_t (\hat{y}_t - y_t)^2, \quad (3)$$

where  $\sigma$  is the training standard deviation and  $\lambda$  sets the emphasis on extremes. The generator minimizes

$$\mathcal{L}_G = -\mathbb{E}[D(\hat{x}, c)] + \beta \mathcal{L}_{\text{recon}}, \quad (4)$$

with  $\beta$  chosen so that the two terms are of comparable magnitude. Training windows are sampled so that the three ENSO phases are equally represented in each mini-batch.

### E. Evaluation Plan

The model is compared with ARIMA, SARIMA and KNN baselines fit on the same observed context, and with ablations that remove the adversarial term, remove the ENSO conditioning, or shuffle the ONI across years. Reported metrics are RMSE, MAE,  $R^2$ , extreme-tail RMSE, and the event-level probability of detection, false alarm ratio and critical success index for heatwave days, over the full test period and separately for El Niño, La Niña and neutral months. Differences in forecast error are tested with the Diebold–Mariano test [21].

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